
LowercaseSuperscripts=false

Final Studies Project Report

Presented in order to obtain

National Diploma of Engineering in Applied Science Technology
speciality : Embedded Systems and Connected Objects

By

Walid Ben Abi Taieb

Comparative assessment of hardware acceleration on the Edge AI

Professional supervisor: Mr. Enrico Mezzetti
Academic supervisor: Mrs. Mariem Feki

Host organisation: Barcelona supercomputing center



Final Studies Project Report

Presented in order to obtain

National Diploma of Engineering in Applied Science Technology
speciality : Embedded Systems and Connected Objects

By

Walid Ben Abi Taieb

Comparative assessment of hardware acceleration on the Edge AI

Professional supervisor : Mr. Enrico Mezzetti

Academic supervisor: Mrs. Mariem Feki

Host organisation: Barcelona supercomputing center



I authorize the student to submit their internship report for validation.

Professional supervisor , **Mr. Enrico Mezzetti**

Signature and stamp

I authorize the student to submit their internship report for validation.

Academic supervisor , **Mrs. Mariem Feki**

Signature

Dedications

Walid Ben Abi Taieb

Acknowledgements

Contents

General introduction	1
1 General context of the project	2
1.1 Barcelona Supercomputing Center (BSC-CNS)	3
1.2 High Performance and Embedded Systems Laboratory (HPES)	3
1.3 Problem statement	4
1.3.1 The computational bottleneck of Edge AI	4
1.3.2 Hardware acceleration as the key enabler	4
1.3.3 The configuration challenge	5
1.3.4 Relevance to this work	5
1.4 Solution approach	6
1.5 Presentation of the proposed solution	6
1.6 Work process and documentation	6
2 Analyse et spécification des besoins	8
2.1 Identification des acteurs du système	9
2.2 Identification des besoins	9
2.2.1 Besoins fonctionnels	9
2.2.2 Besoins non fonctionnels	9
2.3 Diagramme des cas d'utilisation	9
2.4 Identification des scénarios	9
2.5 Diagramme de GANTT théorique	9
3 Conception et choix technologiques	10
3.1 Conception détaillée	11
3.1.1 Diagramme des classes	11
3.1.2 Diagramme des séquences	11
3.2 Conception architecturale	11
3.3 Choix technologiques	11

4 Réalisation	12
4.1 Choix techniques	13
4.2 Travail réalisé	13
4.3 Planning réel du projet	13
Conclusion générale	14
Bibliography	15
Annexes	16
Annexe 1. Exemple d'annexe	16
Annexe 2. Entreprise	17

List of Figures

1.1	Logo of the Barcelona Supercomputing Center.	3
1.2	Logo of the High Performance and Embedded Systems Laboratory (HPES).	4
Annexe 2.1	Logo d'entreprise	17

List of Tables

Annexe 1.1 Exemple tableau dans l'annexe 16

Liste des abréviations

- **GISI** = Génie Informatique des Systèmes Industriels
- **GLSI** = Génie Logiciel et Systèmes d'Information
- **GTR** = Génie des Télécommunications et Réseaux

General introduction

Exemple d'utilisation de la bibliographie utilisée [1]. Le style utilisé est IEEE [2].

Une introduction dure à 3 pages où vous poserez clairement le problème auquel vous allez tenter d'apporter une solution. L'introduction se rédige à la fin de votre travail de rédaction. Avant de rédiger l'introduction, structurez TOUT le PFE. L'introduction peut se faire en même temps que la conclusion.

L'introduction sert trois objectifs :

- elle introduit le sujet. Ceci signifie qu'il faut présenter succinctement le contexte général du travail accompli, par exemple l'environnement professionnel et l'entreprise pour un rapport de stage, puis définir le sujet en termes précis et concis;
- elle énonce ensuite succinctement les objectifs du travail personnel, et les moyens mis en œuvre pour tenter de les atteindre;
- elle s'achève sur une présentation claire du plan adopté pour la suite du corps du rapport. L'annonce du plan se fait au futur et doit être rédigée en entier.

L'introduction générale doit développer les points suivants :

- la présentation du contexte du projet (domaine exemple : télécommunication, sécurité, automate etc.);
- la présentation brève de l'entreprise d'accueil et de son domaine;
- la description des objectifs du PFE/ Mémoire : justifier le sujet et poser le problème à résoudre; indiquer la manière dont il sera traité en termes de outils et de méthodes; donner les raisons qui président à ce choix; exposer les intérêts du sujet et sa problématique;
- l'annonce du plan du rapport sans trop détailler. Il est recommandé, à partir de l'introduction générale, de recourir au *résumé* de modeste.

General context of the project

Contents

1	Barcelona Supercomputing Center (BSC-CNS)	3
2	High Performance and Embedded Systems Laboratory (HPES)	3
3	Problem statement	4
4	Solution approach	6
5	Presentation of the proposed solution	6
6	Work process and documentation	6

Introduction

This introductory chapter presents the host organization and its diverse areas of activity. It then formulates the problem statement and outlines the proposed solution. Finally, the chapter describes the project methodology and provides insights into the adopted approach.

1.1 Barcelona Supercomputing Center (BSC-CNS)

The Barcelona Supercomputing Center (BSC-CNS) is Spain's national supercomputing facility, located in Barcelona. It hosts MareNostrum, one of the most powerful supercomputers in Europe, and serves as a leading research centre in high performance computing (HPC), computer sciences, and computational engineering. BSC-CNS is a member of the Partnership for Advanced Computing in Europe (PRACE) and actively contributes to European research initiatives. Its mission includes providing HPC resources, developing advanced software and tools, and fostering technology transfer to both industry and academia. Figure 1.1 shows the official logo of BSC-CNS.

The centre is organised into several departments, among which the Computer Sciences (CS) department and the Earth Sciences department are particularly prominent. The work presented in this report was carried out within the **High Performance and Embedded Systems Laboratory (HPES)**, which belongs to the Computer Sciences department of BSC-CNS.



Figure 1.1: Logo of the Barcelona Supercomputing Center.

1.2 High Performance and Embedded Systems Laboratory (HPES)

The High Performance and Embedded Systems Laboratory (HPES) at BSC-CNS focuses on the design, analysis, and optimisation of computing systems ranging from large scale parallel machines to embedded real time devices. Key research areas include:

- Parallel programming models and runtime systems.
- Performance analysis and tuning for HPC applications.
- Energy efficient and embedded architectures.
- Fault tolerance and reliability in extreme scale systems.

HPES brings together experts in computer architecture, compilers, operating systems, and application development. The laboratory collaborates on European projects (e.g., EPI, EuroHPC) and with industrial partners to transfer its innovations into practice. Figure 1.2 presents the HPES lab logo.



Figure 1.2: Logo of the High Performance and Embedded Systems Laboratory (HPES).

The present work was conducted within the **SODES** group of HPES, which specialises in the design and optimisation of embedded systems for edge computing and autonomous devices.

1.3 Problem statement

The rapid proliferation of intelligent edge devices from autonomous robots and drones to smart cameras and industrial sensors has created an unprecedented demand for on-device artificial intelligence (Edge AI). Unlike cloud-based AI, Edge AI requires low latency, bandwidth efficiency, data privacy, and energy autonomy. These constraints push computational tasks away from centralised data centres towards resource constrained embedded platforms.

1.3.1 The computational bottleneck of Edge AI

Modern AI models, particularly deep neural networks (DNNs), are computationally intensive. Running them on general purpose CPUs leads to excessive inference times and high energy consumption, often violating real time requirements. While GPUs offer high throughput, their power profile (typically tens to hundreds of watts) makes them unsuitable for battery powered or passively cooled edge devices.

1.3.2 Hardware acceleration as the key enabler

To address this gap, system on chip (SoC) vendors have begun integrating dedicated Neural Processing Units (NPUs) which are specialised accelerators designed to execute tensor operations, convolutions, and activation functions with extreme efficiency. An NPU can deliver 10 to 100 times better performance per watt than a CPU for typical DNN workloads. Examples include the Ethos

U series from Arm, the NPU inside the Rockchip RK3588, and the Neural Engine in Apples A series chips.

1.3.3 The configuration challenge

However, leveraging an NPU is not a matter of simple integration. Each NPU has unique constraints: tiling of tensors into a small local memory (scratchpad), data layout (NHWC vs. NCHW), quantisation schemes (int8, int16, or binary), and supported operations (e.g., depthwise convolutions, activations). Furthermore, the overall system performance depends on the interplay between the NPU, the CPU (for control and preprocessing), the memory hierarchy, and the communication bus.

The core problem is therefore:

How can we determine the optimal configuration (operator partitioning, memory tiling, quantisation, and task scheduling) of a dedicated NPU based heterogeneous system to maximise the acceleration of Edge AI workloads compared to a CPU only baseline, while meeting real time and energy constraints?

1.3.4 Relevance to this work

Within the SODES group at HPES (BSC-CNS), this problem is particularly relevant. Our target embedded platform integrates a CPU cluster and a dedicated NPU. Without proper configuration, the NPU may be underutilised or even slower than a well optimised CPU implementation due to data transfer overhead or mismatched tensor layouts.

This work aims to:

- Characterise the performance gap between CPU only and NPU accelerated execution for representative Edge AI models (object detection, classification models, etc . . .).
- Develop a systematic methodology to explore the configuration space (tiling sizes, quantisation, , DMA settings) and identify Pareto optimal configurations based on the different approaches and standards existing in this field such as The Benchmarking Standard: The MLPerf Tiny for exemple.
- Implement and validate the configuration on real hardware, measuring end to end latency, energy, and throughput.

By solving this configuration problem, we enable practical Edge AI deployment that fully exploits the NPUs potential, turning the promise of hardware acceleration into measurable system level gains.

1.4 Solution approach

The project follows a two stage implementation strategy. First, an initial implementation is carried out on top of the QEMU emulator. This allows us to evaluate the performance of the system across different simulated platforms and to develop a physical model that estimates the latency and throughput of CPU based inference. The emulation on an x86 host provides a controlled environment for early validation.

Second, the emulation is ported to the real hardware platform. This step yields actual performance measurements under realistic operating conditions. Based on these measurements, various optimisation techniques for NPU inference latency are applied. The goal is to identify the configuration that achieves the optimal performance of which the platform is capable.

1.5 Presentation of the proposed solution

The proposed work is built upon successive implementations of the target system. Initially, we simulate the platform using the QEMU emulator. This first implementation serves two purposes: it provides a baseline for CPU based inference through physical emulation, and it enables the development of a performance model that computes the system state and estimates latency and throughput.

Subsequently, we port the system to the actual embedded platform. Inference is executed on the real hardware, allowing a comparative analysis between the general purpose computing units (CPU) and the dedicated machine learning accelerator (NPU). The comparison focuses on inference latency, energy consumption, and throughput, thereby quantifying the benefits of NPU acceleration.

1.6 Work process and documentation

The project follows a structured workflow comprising requirement analysis, iterative development, experimental validation, and documentation. Each phase is documented in the corresponding chapters of this report. Version control is maintained using Git, and experimental results are logged systematically to ensure reproducibility. A detailed project schedule is provided in Appendix A (if

applicable).

Conclusion

This chapter has presented the host organisation and the HPES laboratory, formulated the problem of NPU configuration for Edge AI, and described the proposed two stage solution approach. The next chapter will delve into the requirements analysis and specification for the project.

Analyse et spécification des besoins

Contents

1	Identification des acteurs du système	9
2	Identification des besoins	9
3	Diagramme des cas d'utilisation	9
4	Identification des scénarios	9
5	Diagramme de GANTT théorique	9

Introduction

Introduction partielle, qui annonce le contenu.

2.1 Identification des acteurs du système

2.2 Identification des besoins

2.2.1 Besoins fonctionnels

Citez les besoins fonctionnels ici...

2.2.2 Besoins non fonctionnels

Citez les besoins non fonctionnels ici...

2.3 Diagramme des cas d'utilisation

2.4 Identification des scénarios

2.5 Diagramme de GANTT théorique

Conclusion

Conclusion partielle ayant pour objectif de synthétiser le chapitre et d'annoncer le chapitre suivant.

Conception et choix technologiques

Contents

1	Conception détaillée	11
2	Conception architecturale	11
3	Choix technologiques	11

Introduction

Introduction partielle, qui annonce le chapitre.

3.1 Conception détaillée

Les diagrammes + descriptions textuelles

3.1.1 Diagramme des classes

3.1.2 Diagramme des séquences

3.2 Conception architecturale

architecture physique...

3.3 Choix technologiques

technologies à utiliser pendant la réalisation...

Conclusion

Conclusion partielle ayant pour objectif de synthétiser le chapitre et d'annoncer le chapitre suivant.

Réalisation

Contents

1	Choix techniques	13
2	Travail réalisé	13
3	Planning réel du projet	13

Introduction

Introduction partielle, qui annonce le contenu.

4.1 Choix techniques

4.2 Travail réalisé

4.3 Planning réel du projet

Conclusion

Conclusion partielle ayant pour objectif de synthétiser le chapitre et d'annoncer le chapitre suivant.

Conclusion générale

Rappel du contexte et de la problématique.

Brève récapitulation du travail réalisé et de la solution proposée.

La taille de la conclusion doit être réduite, une page de texte tout au plus. Il est important de souligner que la conclusion ne comporte pas de nouveaux résultats ni de nouvelles interprétations.

Le plus souvent, la conclusion comporte:

- un résumé très rapide du corps du texte;
- un rappel des objectifs du projet;
- un bilan professionnel qui indique clairement les objectifs annoncés dans l'introduction et en particulier ceux qui n'ont pu être atteints. Il présente et synthétise les conclusions partielles;
- un bilan personnel qui décrit les principales leçons que vous tirez de cette expérience sur le plan humain;
- les limites et les perspectives du travail effectué.

Bibliography

- [1] N. Auteur. “Titre de l’article”. [Accès le 19-Janvier-2016], Organisme. [Online]. Available: `http://www.exemple-lien.org/article/?id\_art=124`
- [2] M. Shell. “IEEEtran homepage”. [Accès le 5-Février-2016]. [Online]. Available: `http://www.michaelshell.org/tex/ieeetran/`

Annexes

Annexe 1. Exemple d'annexe

Les chapitres doivent présenter l'essentiel du travail. Certaines informations-trop détaillées ou constituant un complément d'information pour toute personne qui désire mieux comprendre ou refaire une expérience décrite dans le document- peuvent être mises au niveau des annexes. Les annexes, **placées après la bibliographie**, doivent donc être numérotées avec des titres (Annexe1, Annexe2, etc.).

Le tableau annexe 1.1 présente un exemple d'un tableau dans l'annexe.

Tableau annexe 1.1: Exemple tableau dans l'annexe

0	0
1	1
2	2
3	3
4	4

Annexe 2. Entreprise

La figure annexe 2.1 présente le logo entreprise.



Figure annexe 2.1: Logo d'entreprise

arabicAbstract

: رِبَاسْترَ تشيوردس

Résumé

frenchAbstract

Mots clés : frenchAbstractKeywords

Abstract

englishAbstract

Keywords : englishAbstractKeywords

info@bsc.es : : +34.93.413.77.20 :

C/Jordi Girona, 29. Nexus II Building. Office 219. 08034 Barcelona **Tél :** +34.93.413.77.20 **Fax :** **Email :** info@bsc.es

isi@isim.rnu.tn : 71 706 698 : 71 706 164 : 2080 2

2, Abou Raihane Bayrouni 2080 l'Ariana **Tél :** 71 706 164 **Fax :** 71 706 698 **Email :** isi@isim.rnu.tn